



Steps for effective cloud migration

Anyone who's been through a move knows that it's a lot of work.

There's emptying shelves, sorting items for packing or donation, carefully boxing everything up, loading the boxes into a moving van,

and then you have to unpack and get everything organized in its proper place once you get to the new place.

Whew!

An organization migrating its physical computing infrastructure to the cloud also requires careful planning and a great amount of effort to ensure a successful move.

Fortunately, third-party cloud service providers like Google Cloud can help make everything easier.

In this video, we'll discuss the process of migrating an on-premises computing network infrastructure to a cloud platform.

You'll learn the steps to follow, cloud data migration strategies, and important factors to consider during migration.

All right, the **first step** is to think about some **key factors**.

These include choosing the right cloud environment for your organization.

Then, think about how much data will be transferred to the cloud.

This is important because large amounts of data can take a long time to move, which can delay operations.

Next, consider how much downtime your organization can deal with during migration.

Obviously, no business wants to shut down their systems any longer than necessary, so this decision should be agreed on by all stakeholders.

The next step is to choose your migration strategy.

Options include rehosting, also called lift and shift, replatforming, repurchasing, refactoring, or retiring.

Let's break those down.

Rehosting is a cloud-migration strategy that involves moving an entire on-premises system to the cloud without changing anything else about the system.

An exact copy of the current setup is created in the cloud, which helps the organization quickly achieve a return on investment as they use

the enhanced efficiency of their operations, the robust and reliable nature of the cloud infrastructure, and the innovative technologies that are built into cloud-based solutions.

Replatforming is a cloud-migration strategy that involves making small changes to the on-premises system once it's migrated to the cloud.

So, the main structure of the system's applications remains the same, but a few things are improved for better performance.

Repurchasing is a cloud-migration strategy that involves moving applications to a new, cloud-based service platform, usually a software-as-a-service platform.

The cloud service will be an all-new experience, so this requires some team member training.

Refactoring is a cloud-migration strategy that involves building all-new applications from scratch and discarding old applications.

This is ideal when organizations need new features, like serverless computing, that their current systems don't have.

Retiring occurs when applications that are no longer useful are turned off.

The next step in the migration process is choosing your migration partner.

As a cloud professional, you'll want to help your organization find a cloud service provider that offers the right

infrastructure for your particular business, that offers valuable services and tools, and that invests in development to keep things fresh.

It's also important to examine the provider's customer support and service level agreement, or SLA, so you get reliable and prompt support.

Obviously, I'm a big fan of what we do here at Google —especially how we help our partners prepare for cloud migration success!

We've got something called the **Google Cloud Adoption Framework**, which helps users assess their organization's readiness to adopt cloud technologies.

This framework acts like a map from current capabilities to their ideal cloud destination.

The Google Cloud Adoption Framework evaluates four themes.

First, 'learn' refers to the quality and scale of an organization's learning programs.

'Lead' describes the level of support from leadership given to an organization's IT department when migrating to Google Cloud.

'Scale' is the extent to which an organization uses cloud-based services and how much automation they need to manage their system.

And 'secure' ensures an organization's ability to protect their cloud environment from unauthorized and inappropriate access.

Google Cloud provides a migration path that also has four phases: assess, plan, deploy, and optimize.

In the 'assess' phase, users perform a thorough review of their existing network infrastructure.

Then, in the 'plan' phase, users set up a basic cloud infrastructure on Google Cloud, where their workloads will exist.

When it's time to 'deploy', the workloads are actually moved to Google Cloud.

Lastly, the 'optimize' phase is when organizations begin using cloud-based technologies and features.

And this is where they start to really enjoy the improved accessibility, scalability, cost-savings, security, and efficiency.

Like any move, cloud migration requires careful planning.

As a cloud data analytics professional, the ability to help your organization follow the steps in this video will help

ensure everything gets to where it needs to be so you can enjoy your beautiful new place in the cloud!